

For the use of a Registered Medical Practitioner or a Hospital or a Laboratory only OR for Specialist Use only

TADACIP 20

Tadalafil Tablets 20 mg

COMPOSITION

Each film-coated tablet contains:

Tadalafil Ph.Eur.20 mg

Colour: Titanium dioxide & Yellow oxide of iron

DOSAGE FORM

Film coated tablets

PRODUCT DESCRIPTION

Yellow coloured, capsule shaped, biconvex film coated tablet with “T 20” debossing on one side and plain on other side.

PHARMACOLOGY

Pharmacodynamics

Pharmacotherapeutic group: Urologicals, Drugs used in erectile dysfunction

ATC Code: G04BE08.

Mechanism of action

Tadalafil is a selective, reversible inhibitor of cyclic guanosine monophosphate (cGMP)-specific phosphodiesterase type 5 (PDE5).

Erectile dysfunction

When sexual stimulation causes the local release of nitric oxide, inhibition of PDE5 by tadalafil produces increased levels of cGMP in the corpus cavernosum. This results in smooth muscle relaxation and inflow of blood into the penile tissues, thereby producing an erection. Tadalafil has no effect in the treatment of erectile dysfunction in the absence of sexual stimulation.

Pulmonary arterial hypertension

Pulmonary arterial hypertension is associated with impaired release of nitric oxide by the vascular endothelium and consequent reduction of cGMP concentrations within the pulmonary vascular smooth muscle. PDE5 is the predominant phosphodiesterase in the pulmonary vasculature. Inhibition of PDE5 by tadalafil increases the concentrations of cGMP resulting in relaxation of the pulmonary vascular smooth muscle cell and vasodilation of the pulmonary vascular bed.

Pharmacodynamic effects

Studies in vitro have shown that tadalafil is a selective inhibitor of PDE5. PDE5 is an enzyme found in corpus cavernosum smooth muscle, vascular and visceral smooth muscle, skeletal muscle, platelets, kidney, lung, and cerebellum. The effect of tadalafil is more potent on PDE5 than on other phosphodiesterases. Tadalafil is > 10,000-fold more potent for PDE5 than for PDE1, PDE2, and PDE4 enzymes which are found in the heart, brain, blood vessels, liver, and other organs. Tadalafil is > 10,000-fold more potent for PDE5 than for PDE3, an enzyme found in the heart and blood vessels.

This selectivity for PDE5 over PDE3 is important because PDE3 is an enzyme involved in cardiac contractility. Additionally, tadalafil is approximately 700-fold more potent for PDE5 than for PDE6, an enzyme which is found in the retina and is responsible for phototransduction. Tadalafil is also > 10,000-fold more potent for PDE5 than for PDE7 through PDE10.

Paediatric population

The overall safety results were generally consistent with the known safety profile of tadalafil and with adverse events (AEs) expected in a paediatric DMD population receiving corticosteroids.

The European Medicines Agency has waived the obligation to submit the results of studies in all subsets of the paediatric population in the treatment of the erectile dysfunction and in one or more subsets of the paediatric population in the treatment of pulmonary arterial hypertension.

Pharmacokinetics

Absorption

Tadalafil is readily absorbed after oral administration and the mean maximum observed plasma concentration (C_{max}) is achieved at a median time of 2 hours after dosing. Absolute bioavailability of tadalafil following oral dosing has not been determined.

The rate and extent of absorption of tadalafil are not influenced by food, thus tadalafil may be taken with or without food. The time of dosing (morning versus evening) had no clinically relevant effects on the rate and extent of absorption.

Distribution

The mean volume of distribution is approximately 63 l, indicating that tadalafil is distributed into tissues. At therapeutic concentrations, 94% of tadalafil in plasma is bound to proteins. Protein binding is not affected by impaired renal function.

Less than 0.0005% of the administered dose appeared in the semen of healthy subjects.

Biotransformation

Tadalafil is predominantly metabolised by the cytochrome P450 (CYP) 3A4 isoform. The major circulating metabolite is the methylcatechol glucuronide. This metabolite is at least 13,000-fold less potent than tadalafil for PDE5. Consequently, it is not expected to be clinically active at observed metabolite concentrations.

Elimination

The mean oral clearance for tadalafil is 2.5 l/h and the mean half-life is 17.5 hours.

Tadalafil is excreted predominantly as inactive metabolites, mainly in the faeces (approximately 61% of the dose) and to a lesser extent in the urine (approximately 36% of the dose).

Linearity/Non-Linearity

Tadalafil pharmacokinetics in healthy subjects are linear with respect to time and dose. Over a dose range of 2.5 to 20 mg, exposure (AUC) increases proportionally with dose. Steady-state plasma concentrations are attained within 5 days of once daily dosing.

Pharmacokinetics determined with a population approach in patients with erectile dysfunction are similar to pharmacokinetics in subjects without erectile dysfunction.

Special Populations

Elderly

Healthy elderly population (65 years or over) had a lower oral clearance of tadalafil, resulting in 25% higher exposure (AUC) relative to healthy population aged 19 to 45 years. This effect of age is not clinically significant and does not warrant a dose adjustment.

Renal Insufficiency

Single dose tadalafil (5 mg-20 mg), tadalafil exposure (AUC) approximately doubled in subjects with mild (creatinine clearance 51 to 80ml/min) or moderate (creatinine clearance 31 to 50ml/min) renal impairment and in subjects with end-stage renal disease on dialysis. In haemodialysis patients, C_{max} was 41% higher than that observed in healthy subjects. Haemodialysis contributes negligibly to tadalafil elimination.

Hepatic Insufficiency

There is limited clinical data on the safety of tadalafil in patients with severe hepatic insufficiency (Child-Pugh class C). If tadalafil is prescribed, a careful individual benefit/risk evaluation should be undertaken by the prescribing physician. There are no available data about the administration of once-a-day dosing of tadalafil to patients with hepatic impairment. If tadalafil is prescribed once-a-day, a careful individual benefit/risk evaluation should be undertaken by the prescribing physician. There are no available data about the administration of doses higher than 10 mg of tadalafil to patients with hepatic impairment.

Patients with Diabetes

Tadalafil exposure (AUC) in patients with diabetes was approximately 19% lower than the AUC value for healthy population. This difference in exposure does not warrant a dose adjustment.

Race

Pharmacokinetic studies have included subjects and patients from different ethnic groups, and no differences in the typical exposure to tadalafil have been identified. No dose adjustment is warranted.

Gender

In healthy female and male subjects following single and multiple-doses of tadalafil, no clinically relevant differences in exposure were observed. No dose adjustment is warranted.

INDICATIONS

Treatment of erectile dysfunction in adult males.

In order for tadalafil to be effective for the treatment of erectile dysfunction, sexual stimulation is required.

Tadalafil is not indicated for use by women.

DOSAGE AND METHOD OF ADMINISTRATION

Posology

Erectile dysfunction in adult Men

In general, the recommended dose is 10 mg taken prior to anticipated sexual activity and with or without food.

In those patients in whom tadalafil 10 mg does not produce an adequate effect, 20 mg might be tried. It may be taken at least 30 minutes prior to sexual activity.

The maximum dose frequency is once per day.

Tadalafil 20 mg is intended for use prior to anticipated sexual activity and it is not recommended for continuous daily use.

In patients who anticipate a frequent use of tadalafil (i.e., at least twice weekly) a once daily regimen with the lowest doses of tadalafil might be considered suitable, based on patient choice and the physician's judgement.

In these patients, the recommended dose is 5mg taken once a day at approximately the same time of day. The dose may be decreased to 2.5mg once a day based on individual tolerability.

The appropriateness of continued use of the daily regimen should be reassessed periodically.

Special Populations

Elderly Men

Dose adjustments are not required in elderly patients.

Men with Renal Impairment

Dose adjustments are not required in patients with mild to moderate renal impairment. For patients with severe renal impairment, 10 mg is the maximum recommended dose for on-demand treatment. (*See Warning and Precaution and Pharmacodynamics.*)

Once-a-day dosing of tadalafil is not recommended in patients with severe renal impairment.

Men with Hepatic Impairment

For the treatment of erectile dysfunction using on-demand tadalafil tablet recommended dose of tadalafil tablet is 10 mg taken prior to anticipated sexual activity and with or without food. There is limited clinical data on the safety of tadalafil in patients with severe hepatic impairment (Child-Pugh class C); if prescribed, a careful individual benefit/risk evaluation should be undertaken by the prescribing physician. There are no available data about the administration of doses higher than 10 mg of tadalafil to patients with hepatic impairment. Once-a-day dosing of tadalafil tablet both for the treatment of erectile dysfunction and benign prostatic hyperplasia has not been evaluated in patients with hepatic impairment; therefore if prescribed, a careful individual benefit/risk evaluation should be undertaken by the prescribing physician.

(See Warning and Precaution and Pharmacodynamics)

Men with Diabetes

Dose adjustments are not required in diabetic patients.

Paediatric population

There is no relevant use of tadalafil in the paediatric population with regard to the treatment of erectile dysfunction.

Method of administration

Tadalafil film-coated tablets are for oral use only.

CONTRAINDICATIONS

Hypersensitivity to the active substance or to any of the excipients.

Tadalafil was shown to augment the hypotensive effects of nitrates. This is thought to result from the combined effects of nitrates and tadalafil on the nitric oxide/cGMP pathway. Therefore, administration of tadalafil to patients who are using any form of organic nitrate is contraindicated. *(See Drug Interaction)*

Tadalafil must not be used in men with cardiac disease for whom sexual activity is inadvisable. Physicians should consider the potential cardiac risk of sexual activity in patients with pre-existing cardiovascular disease.

The use of tadalafil is contraindicated following groups of patient with cardiovascular disease:

- Patients with myocardial infarction within the last 90 days.
- Patients with unstable angina or angina occurring during sexual intercourse.
- Patients with New York Heart Association class 2 or greater heart failure in the last 6 months.
- Patients with uncontrolled arrhythmias, hypotension (<90/50mmHg), or uncontrolled hypertension.
- Patients with a stroke within the last 6 months.

Tadalafil is contraindicated in patients who have loss of vision in one eye because of non-arteritic anterior ischaemic optic neuropathy (NAION), regardless of whether this episode was in connection or not with previous PDE5 inhibitor exposure (*see Warning and Precaution*).

The co-administration of PDE5 inhibitors, including tadalafil, with guanylate cyclase stimulators, such as riociguat, is contraindicated as it may potentially lead to symptomatic hypotension (*see Drug Interactions*).

WARNINGS AND PRECAUTIONS

Before treatment with Tadalafil

A medical history and physical examination should be undertaken to diagnose erectile dysfunction and determine potential underlying causes, before pharmacological treatment is considered.

Prior to initiating any treatment for erectile dysfunction, physicians should consider the cardiovascular status of their patients, since there is a degree of cardiac risk associated with sexual activity. Tadalafil has vasodilator properties, resulting in mild and transient decreases in blood pressure, and as such potentiates the hypotensive effect of nitrates.

The evaluation of erectile dysfunction should include a determination of potential underlying causes and the identification of appropriate treatment following an appropriate medical assessment. It is not known if tadalafil is effective in patients who have undergone pelvic surgery or radical non-nerve-sparing prostatectomy.

Cardiovascular

Serious cardiovascular events, including myocardial infarction, sudden cardiac death, unstable angina pectoris, ventricular arrhythmia, stroke, transient ischaemic attacks, chest pain, palpitations and tachycardia, have been reported either post marketing and/or in clinical trials. Most of the patients in whom these events have been reported had pre-existing cardiovascular risk factors. However, it is not possible to definitively determine whether these events are related directly to these risk factors, to tadalafil, to sexual activity, or to a combination of these or other factors.

The following groups of patients with cardiovascular disease were not included in PAH clinical studies:

- Patients with clinically significant aortic and mitral valve disease
- Patients with pericardial constriction
- Patients with restrictive or congestive cardiomyopathy
- Patients with significant left ventricular dysfunction
- Patients with life-threatening arrhythmias
- Patients with symptomatic coronary artery disease
- Patients with uncontrolled hypertension.

Since there are no clinical data on the safety of tadalafil in these patients, the use of tadalafil is not recommended.

Pulmonary vasodilators may significantly worsen the cardiovascular status of patients with pulmonary veno-occlusive disease (PVOD). Since there are no clinical data on administration of tadalafil to patients with veno-occlusive disease, administration of tadalafil to such patients is not recommended. Should signs of pulmonary oedema occur when tadalafil is administered, the possibility of associated PVOD should be considered.

Tadalafil has systemic vasodilatory properties that may result in transient decreases in blood pressure. Physicians should carefully consider whether their patients with certain underlying conditions, such as severe left ventricular outflow obstruction, fluid depletion, autonomic hypotension or patients with resting hypotension, could be adversely affected by such vasodilatory effects.

In patients who are taking alpha1 blockers, concomitant administration of tadalafil may lead to symptomatic hypotension in some patients. The combination of tadalafil and doxazosin is not recommended.

Vision

Visual defects and cases of NAION have been reported in connection with the intake of tadalafil and other PDE5 inhibitors. Analyses of observational data suggest an increased risk of acute NAION in men with erectile dysfunction following exposure to tadalafil or other PDE5 inhibitors. As this may be relevant for all patients exposed to tadalafil, the patient should be advised that in case of sudden visual defect, he should stop taking Tadalafil and consult a physician immediately.

Decreased or sudden hearing loss

Cases of sudden hearing loss have been reported after the use of tadalafil. Although other risk factors were present in some cases (such as age, diabetes, hypertension and previous hearing loss history) patients should be advised to stop taking tadalafil and seek prompt medical attention in the event of sudden decrease or loss of hearing.

Renal and hepatic impairment

Due to increased tadalafil exposure (AUC), limited clinical experience and the lack of ability to influence clearance by dialysis, once-a-day dosing of Tadalafil is not recommended in patients with severe renal impairment.

There is limited clinical data on the safety of single-dose administration of tadalafil in patients with severe hepatic insufficiency (Child-Pugh class C). If Tadalafil is prescribed, a careful individual benefit/risk evaluation should be undertaken by the prescribing physician.

Priapism and anatomical deformation of the penis

Patients who experience erections lasting 4 hours or more should be instructed to seek immediate medical assistance. If priapism is not treated immediately, penile tissue damage and permanent loss of potency may result.

Tadalafil should be used with caution in patients with anatomical deformation of the penis (such as angulation, cavernosal fibrosis, or Peyronie's disease) or in patients who have conditions which may predispose them to priapism (such as sickle cell anaemia, multiple myeloma, or leukaemia).

Use with CYP3A4 inducers or inhibitors

Caution should be exercised when prescribing Tadalafil to patients using potent CYP3A4 inhibitors (ritonavir, saquinavir, ketoconazole, itraconazole, and erythromycin), as increased tadalafil exposure (AUC) has been observed if the medicinal products are combined.

Tadalafil and other treatments for erectile dysfunction

The safety and efficacy of combinations of tadalafil and other PDE5 inhibitors or other treatments for erectile dysfunction have not been studied. The patients should be informed not to take Tadalafil in such combinations.

Prostacyclin and its analogues

The efficacy and safety of tadalafil co-administered with prostacyclin or its analogues has not been studied in controlled clinical studies. Therefore, caution is recommended in case of co-administration.

Bosentan

The efficacy of tadalafil in patients already on bosentan therapy has not been conclusively demonstrated.

Lactose

Tadalafil contains lactose. Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicinal product.

DRUG INTERACTIONS

Effects of Other Substances on Tadalafil

Cytochrome P450 inhibitors: Tadalafil is principally metabolised by CYP3A4. CYP3A4 inhibitors, such as ketoconazole, erythromycin, clarithromycin, itraconazole, and grapefruit juice, and protease inhibitors, such as ritonavir, saquinavir should be co-administered with caution, as they would be expected to increase plasma concentrations of tadalafil.

Transporters: The role of transporters (for example, p-glycoprotein) in the disposition of tadalafil is not known. Therefore, there is the potential of drug interactions mediated by inhibition of transporters.

Cytochrome P450 inducers: CYP3A4 inducer, such as rifampicin, phenobarbital, phenytoin, and carbamazepine may decrease plasma concentrations of tadalafil, thus anticipated to decrease the efficacy of tadalafil; the magnitude of decreased efficacy is unknown.

Effects of Tadalafil on Other Medicinal Products

Nitrates: Administration of Tadalafil tablet to patients who are using any form of organic nitrate is contraindicated, as tadalafil was shown to augment the hypotensive effects of nitrates.

Anti-hypertensives (including calcium channel blockers): Caution should be exercised when using tadalafil in patients treated with any alpha-blockers (such as doxazosin, alfuzosin or tamsulosin), and notably in the elderly. Treatments should be initiated at minimal dosage and progressively adjusted. Appropriate clinical advice should be given to patients regarding a possible decrease in blood pressure when they are treated with antihypertensive medicinal products.

5- α reductase inhibitors: Caution should be exercised when tadalafil is co-administered with 5- α reductase inhibitors (5-ARIs), as a formal drug-drug interaction study evaluating the effects of tadalafil and 5-ARIs has not been performed.

CYP1A2 substrates (e.g. theophylline): It should be considered when co-administering theophylline (a non-selective phosphodiesterase inhibitor) with Tadalafil tablet.

Ethinylestradiol and terbutaline: Tadalafil has been demonstrated to produce an increase in the oral bioavailability of ethinylestradiol; a similar increase may be expected with oral administration of terbutaline, although the clinical consequence of this is uncertain.

Alcohol: The effect of alcohol on cognitive function was not augmented by tadalafil.

Cytochrome P450 metabolised medicinal products: Tadalafil is not expected to cause clinically significant inhibition or induction of the clearance of medicinal products metabolised by CYP450 isoforms. Studies have confirmed that tadalafil does not inhibit or induce CYP450 isoforms, including CYP3A4, CYP1A2, CYP2D6, CYP2E1, CYP2C9 and CYP2C19.

CYP2C9 substrates (e.g. R-warfarin): Tadalafil had no clinically significant effect on exposure (AUC) to S-warfarin or R-warfarin (CYP2C9 substrate), nor did tadalafil affect changes in prothrombin time induced by warfarin.

Aspirin: Tadalafil did not potentiate the increase in bleeding time caused by acetylsalicylic acid.

Antidiabetic medicinal products: Specific interaction studies with antidiabetic medicinal products were not conducted.

FERTILITY, PREGNANCY AND LACTATION

Tadalafil is not indicated for use by women.

Pregnancy

There are limited information from the use of tadalafil in pregnant women. Animal studies do not indicate direct or indirect harmful effects with respect to pregnancy, embryonal/foetal development, parturition or postnatal development. As a precautionary measure, it is preferable to avoid the use of tadalafil during pregnancy.

Lactation

Available pharmacodynamic/toxicological data in animals have shown excretion of tadalafil in milk. A risk to the suckling child cannot be excluded. Tadalafil should not be used during breast feeding.

Fertility

Effects were seen in dogs that might indicate impairment of fertility. This effect is unlikely in humans, although a decrease in sperm concentration was seen in some men (*see Pharmacodynamics*)

UNDESIRABLE EFFECTS

Summary of the safety profile

Very common	Common	Uncommon	Rare
<i>Immune system disorders</i>			
		Hypersensitivity reactions	Angioedema ²
<i>Nervous system disorders</i>			
	Headache	Dizziness	Stroke ¹ (including haemorrhagic events), Syncope, Transient

			ischaemic attacks ¹ , Migraine ² , Seizures ² , Transient amnesia
<i>Eye disorders</i>			
		Blurred vision, Sensations described as eye pain	Visual field defect, Swelling of eyelids, Conjunctival hyperaemia, Non-arteritic anterior ischaemic optic neuropathy (NAION) ² , Retinal vascular occlusion ²
<i>Ear and labyrinth disorders</i>			
		Tinnitus	Sudden hearing loss
<i>Cardiac disorders¹</i>			
		Tachycardia, Palpitations	Myocardial infarction, Unstable angina pectoris ² , Ventricular arrhythmia ²
<i>Vascular disorders</i>			
	Flushing	Hypotension ³ , Hypertension	
<i>Respiratory, thoracic and mediastinal disorders</i>			
	Nasal congestion	Dyspnoea, Epistaxis	
<i>Gastrointestinal disorders</i>			
	Dyspepsia	Abdominal pain, Vomiting, Nausea, Gastro- oesophageal reflux	
<i>Skin and subcutaneous tissue disorders</i>			
		Rash	Urticaria, Stevens-Johnson syndrome ² , Exfoliative dermatitis ² , Hyperhydrosis (sweating)
<i>Musculoskeletal, connective tissue and bone disorders</i>			
	Back pain, Myalgia, Pain in extremity		

<i>Renal and urinary disorders</i>			
		Haematuria	
<i>Reproductive system and breast disorders</i>			
		Prolonged erections	Priapism, Penile haemorrhage, Haemospermia
<i>General disorders and administration site conditions</i>			
		Chest pain ¹ , Peripheral oedema, Fatigue	Facial oedema ² , Sudden cardiac death ^{1,2}

- (1) Most of the patients had pre-existing cardiovascular risk factors (*see Warnings and Precautions*).
- (2) Postmarketing surveillance reported adverse reactions not observed in placebo-controlled clinical trials.
- (3) More commonly reported when tadalafil is given to patients who are already taking antihypertensive medicinal products.

Description of selected adverse reactions

A slightly higher incidence of ECG abnormalities, primarily sinus bradycardia, has been reported in patients treated with tadalafil once a day as compared with placebo. Most of these ECG abnormalities were not associated with adverse reactions.

Other special populations

Data in patients over 65 years of age receiving tadalafil in clinical trials, either for the treatment of erectile dysfunction or the treatment of benign prostatic hyperplasia, are limited. In clinical trials with tadalafil taken on demand for the treatment of erectile dysfunction, diarrhoea was reported more frequently in patients over 65 years of age. In clinical trials with tadalafil 5 mg taken once a day for the treatment of benign prostatic hyperplasia, dizziness and diarrhoea were reported more frequently in patients over 75 years of age.

Reporting of suspected adverse reactions

Reporting suspected adverse reactions after authorisation of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product. Healthcare professionals are asked to report any suspected adverse reactions via the local reporting system

OVERDOSE

Single doses of up to 500 mg have been given to healthy people, and multiple daily doses up to 100 mg have been given to patients. Adverse events were similar to those seen at lower doses.

In cases of overdose, standard supportive measures should be adopted, as required. Haemodialysis contributes negligibly to tadalafil elimination.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINE

Tadalafil has negligible influence on the ability to drive or use machines. Although the frequency of reports of dizziness in placebo and tadalafil arms was similar, patients should be aware of how they react to tadalafil before driving and use machine.

INCOMPATIBILITIES

Not applicable.

STORAGE AND HANDLING INSTRUCTION

Store below 30 °C.

Store in original packaging.

SHELF LIFE

24 Months

PACKAGING INFORMATION

Carton containing ALU-PVC/PE/PVDC blister of 1 x 4 tablets each.

Carton containing ALU-PVC/PE/PVDC blister of 2 x 4 tablets each.

PRODUCT REGISTRATION HOLDER

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MANUFACTURER

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Verna Industrial Estate

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